

Report

On

Energy-Efficient Gateway Placement and Resilient Routing in IoT Networks

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Chapter 1

Introduction

1.1 Background

IoT deployments frequently use low-power sensor nodes to monitor physical environ-ments and forward measurements to gateways. A gateway acts as an aggregation and forwarding point between local sensor devices and a wider network. If gateways are poorly located, sensors may require longer transmissions, increasing energy use and potentially causing early battery depletion. Conversely, increasing the number of gateways can reduce average radio distance but also raises infrastructure cost.

Wireless sensor networks are energy constrained, so gateway placement should be evaluated with more than geometric distance alone. Resilience is also important: when a gateway fails, a sensor should ideally have another in-range gateway or a viable relay path. This project implements a compact simulator for exploring these interacting factors.

1.2 Problem statement

In an IoT network, battery-powered sensor nodes send data to a gateway, and the energy needed grows quickly with distance. A poorly placed gateway drains batteries early, raises maintenance cost and can leave parts of the network without service. Real sensor deployments are uneven and often grouped in dense clusters, so random placement is rarely a good choice. Adding more gateways helps but increases cost, and one gateway failure can cut off many nodes at once. This project compares random and K-Means gateway placement in a simulated field of 200 nodes, using distance, radio energy and node lifetime. It then examines how gateway count, backup gateways and multi-hop forwarding affect energy use, reliability and cost.

1.3 Objectives

1. Implement a repeatable two-dimensional sensor deployment model.
2. Compare random and K-Means gateway placement for $k = 1, \ldots, 8$.
3. Estimate expected and full-load transmission energy using a first-order radio model.
4. Estimate first-node-death (FND) and 50%-node-death rounds under stochastic traffic.
5. Explore gateway-count selection using the elbow method, silhouette score, and an illustrative cost score.
6. Evaluate a simplified single-gateway-failure model with backup and spare gateway options.
7. Compare direct routing with minimum-energy and residual-energy-aware multi-hop routing.
8. Provide CSV outputs, figures, a browser dashboard, and a validation script.

1.4 Scope and boundaries

This project is a software simulation only. It does not use physical sensors or radio hardware, and it does not rely on field measurements, captured network packets or a real deployment. The 200 sensor nodes are generated synthetically, so the results show how different design choices behave under the stated assumptions, not how a specific real network would perform. Effects such as signal fading, interference and packet loss are not modelled. All the numbers in the result tables come from the Python scripts, which are the main source of evidence in this report. The browser dashboard is a separate, exploratory tool meant for live demonstration, and because it has its own random layouts and simplified settings, its numbers may not exactly match the tables. The base placement experiment does not apply the 80 m communication range limit, while the extended fault-tolerance and routing experiments do. For this reason, the results of these two parts should be read separately and not compared directly with each other.

Chapter 2

Related Concepts

2.1 Gateway Placement

Gateway placement is an important part of designing an energy-efficient Wireless Sensor Net-work (WSN). Gateways collect data from sensor nodes and forward it to the wider network. Their locations affect communication distance, energy consumption, and infrastructure cost. Random placement provides a simple baseline, but gateways may be far from densely popu-lated sensor regions. In comparison, K-Means groups sensor nodes into k clusters and places a gateway at each cluster centre. Although this approach reduces the within-cluster squared distance, it does not directly optimize energy consumption, reliability, or total cost .

2.2 Radio-Energy Model

Sensor nodes usually operate with limited battery power, so communication energy is an im-portant factor in WSN performance. This project uses a first-order radio-energy model that considers both electronic circuit energy and the additional energy required to transmit data over a distance. According to this model, longer transmission distances generally require more energy. The estimated results therefore depend on the selected model parameters .

2.3 Cluster-Count Selection

Selecting the number of gateways is important because more gateways can reduce communi-cation distances but increase deployment cost. The elbow method examines how K-Means inertia decreases as the number of clusters increases. The silhouette coefficient measures how well sensor nodes fit within their assigned clusters compared with neighbouring clusters [3]. These measures help compare possible gateway counts, but the final choice should a

consider energy consumption, coverage, reliability, and cost.

2.4 Shortest-Path Routing

When a sensor cannot communicate directly with a gateway, intermediate nodes may help forward its data. This project uses a Dijkstra-style shortest-path algorithm to identify routes through available relay links [4]. The routing cost is based on the radio-energy model, while the energy-aware approach also considers the remaining battery power of relay nodes. This helps explore how routing decisions affect energy usage and network lifetime. However, the model does not fully represent interference, congestion, packet loss, or other practical communication challenges.

2.5 IOT Network Architecture

An IoT network usually has three layers. The sensing layer contains battery-powered sensor nodes that measure things such as temperature, soil moisture or air quality. The gateway layer collects data from nearby nodes and forwards it onward. The cloud or application layer stores the data and shows it to users through dashboards. Sensor nodes are small and have limited battery, memory and processing power, so they are called constrained devices. Gateways act as the bridge between these constrained nodes and the internet, which is why their position and number matter so much for the whole network.

2.6 K-Means Clustering

K-Means is a basic unsupervised learning method that divides a set of points into k groups. It starts with k initial centres, assigns every point to its nearest centre, and then moves each centre to the average position of its group. These two steps repeat until the centres stop changing. In this project the points are sensor positions, and the final cluster centres are used as gateway locations. Since the result can depend on the starting centres, the algorithm is run several times and the best result is kept.

Chapter 3

Implementation

3.1 Software architecture

The project contains the following components:

- gateway_placement.py: base placement, energy, lifetime, and three plots.
- gateway_extended.py: cluster-count analysis, fault tolerance, routing, and three plots.
- gateway_sim.html: interactive browser-based visual exploration.
- results.csv and extended_results.csv: numerical outputs.
- validate_project.py: syntax, runtime, and output-file checks.

3.2 Deployment model

The simulated field is a square with a side length of 200 m, and the base experiment places N = 200 sensor nodes inside it. Real sensors are rarely spread out evenly, so the deployment is made non-uniform on purpose. Seventy percent of the nodes are placed around three randomly chosen cluster centres, with a standard deviation of 15 m, which creates three dense groups similar to machine clusters or farm sections. The remaining thirty percent are placed uniformly across the field as scattered background sensors, and any coordinate outside the field is clipped to its boundary. The main random seed is fixed at 42, so the same layout is generated every time the script is run. K-Means is run with ten initializations and the best result is kept, which avoids a poor clustering caused by an unlucky start. Because random placement depends heavily on luck, the random baseline is averaged over 30 placement trials for each tested gateway count. Each node connects to its nearest gateway.

3.3 Radio-energy equations

For a packet of L bits transmitted over distance d, the model is

$$E_{\mathrm{TX}}(L,d) = L\left(E_{\mathrm{elec}} + \epsilon_{\mathrm{amp}} d^2\right) \tag{3.1}$$

where E_{elec} is the electronics energy per bit and ϵ_{amp} is the amplifier coefficient. The receiver energy is

$$E_{\mathrm{RX}}(L) = LE_{\mathrm{elec}}. \tag{3.2}$$

The configured parameters are listed in Table 3.1.

Table 3.1: Principal simulation parameters

Parameter	Value
Field dimensions	200 m × 200 m
Number of sensor nodes	200
Gateway counts (base)	$k = 1, 2, \ldots, 8$
Packet length, L	4000 bits
Electronics energy, E_{elec}	50 nJ/bit
Amplifier coefficient, ϵ_{amp}	100 pJ/(bit m^2)
Initial sensor battery, E_0	0.5 J
Per-round transmission probability	0.5
Base maximum rounds	200000
Extended direct range	80 m
Extended relay-link range	40 m
Extended routing gateway count	2
Extended maximum rounds	8000

For a set of nodes with assigned gateway distances d_i, the full-load energy reference is

$$E_{\mathrm{full}} = \sum_{i=1}^{N} E_{\mathrm{TX}}(L, d_i) \tag{3.3}$$

and the expected sensor transmission energy per round under independent transmission probability p_{tx} is

$$E_{\mathrm{expected}} = p_{\mathrm{tx}} \sum_{i=1}^{N} E_{\mathrm{TX}}(L, d_i) \tag{3.4}$$

These quantities are not interchangeable. The expected energy in Equation 3.4 accounts

for the probability that a node transmits; the full-load reference assumes all nodes transmit in every round. The reported base energy metric covers sensor-to-gateway transmission energy, not a complete network-wide accounting of all possible gateway, receive, processing, and backhaul costs.

3.4 Placement methods

3.4.1 Random placement

For each gateway count k, gateway coordinates are sampled uniformly within the field. Every sensor is assigned to its nearest gateway. Thirty placements are evaluated and the reported random baseline is the mean of the resulting metrics. The base experiment does not reject assignments whose distance exceeds 80 m.

3.4.2 K-Means placement

K-Means is fitted to the sensor coordinates for each value of k. The k cluster centres are treated as gateway locations. Each node is assigned to its nearest centre. This reduces within-cluster squared distance for the fitted data, but does not explicitly optimize fault tolerance or the full network energy objective.

3.5 Lifetime metrics

Each sensor starts with $E_0 = 0.5$ J. At each round, an alive node transmits with probability $p_{tx} = 0.5$ and loses the energy associated with its assigned link. Dead nodes are not permitted to transmit. The first-node-death metric is

$$\mathrm{FND} = \min\{r : \text{at least one node has } E_i(r) \le 0\}, \tag{3.5}$$

and the half-dead metric is the first round in which at least half the nodes

have depleted their battery. The lifetime experiment is stochastic; the fixed seed supports repeatability of the included run, but does not establish statistical significance.

3.6 Gateway-count and cost analysis

The extended experiment evaluates $k = 1$ to 10 using K-Means inertia, silhouette score, and a cost score. The cost model uses the assumptions C_g = 200 arbitrary cost units per

gateway, $C_b = 5$ arbitrary units per battery pack, horizon $H = 5000$ rounds, and E_0 = 0.5

J. Its battery-equivalent term is implemented as

$$B(k) = \frac{E_{\text{ref}}(k)\, p_{\text{tx}}\, H}{E_0}, \tag{3.6}$$

with total score

$$C_{\text{total}}(k) = kC_g + B(k)C_b. \tag{3.7}$$

Here, $E_{\text{ref}}(k)$ is the sum of per-packet transmission energies used by the code for the can-didate gateway count. This is an illustrative battery-equivalent cost score, not a currency-denominated procurement estimate. It excludes installation, maintenance, gateway power, backhaul, battery replacement logistics, and discounting.

3.7 Fault-tolerance model

The extended experiment uses an 80 m sensor-to-gateway communication range. A sensor can deliver after a gateway failure if an operational gateway is within this range and the con-figured primary/backup rule permits it. The script computes the worst-case delivery ratio over single-gateway failures. It compares K-Means with no backup, K-Means with backup selection, a random-placement backup baseline, and K-Means plus one additional gateway placed by a grid search over candidate positions. This metric is an in-range connectivity proxy; it does not model failure detection delay, packet loss, congestion, or gateway repair.

3.8 Multi-hop routing model

The routing experiment uses two K-Means gateways, an 80 m direct communication range, and a 40 m relay-link range. It compares three modes:

1. **Direct:** each node sends directly to a gateway when an in-range gateway is available.
2. **Minimum-energy multi-hop:** Dijkstra-style route construction uses radio energy and relay receive energy as path-cost components.
3. **Residual-energy-aware multi-hop:** the route cost is weighted by a function of re-maining battery, intended to penalize routes that burden low-energy nodes.

The simulation uses a synchronous-round simplification: packets for a round are processed before depleted nodes are marked unavailable at the round boundary. Routes are periodi-cally recomputed and may be refreshed after node depletion. This timing assumption should be kept in mind when interpreting lifetime values.

Chapter 4

Results

4.1 Placement comparison

Figure 4.1 shows the example spatial comparison between random and K-Means gateway locations at $k = 4$. The base CSV results for all gateway counts are reported in Table 4.1. Values are from the included reproducible simulation output; energy values in the table are expected energy per round, not full-load energy.

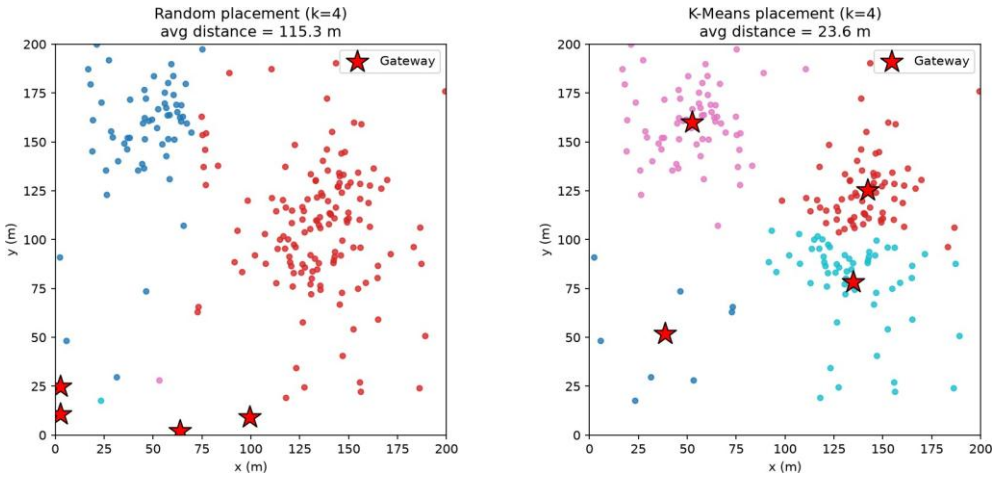


Figure 4.1: Example random and K-Means gateway placement for $k = 4$.

Table 4.1: Base placement results: expected energy and lifetime

k	Method	Avg. distance (m)	Expected energy (mJ/round)	FND (rounds)	50% dead (rounds)
1	Random	95.962	473.020	70	361

Continued on next page

Table 4.1 – continued from previous page

k	Method	Avg. distance (m)	Expected energy (mJ/round)	FND (rounds)	50% dead (rounds)
1	K-Means	55.697	171.320	150	805
2	Random	69.340	263.727	104	602
2	K-Means	31.637	80.878	139	2151
3	Random	57.722	198.248	132	855
3	K-Means	26.307	62.681	181	2515
4	Random	52.518	162.505	178	970
4	K-Means	23.560	50.920	388	2734
5	Random	43.023	115.292	203	1226
5	K-Means	21.408	44.783	484	2907
6	Random	43.838	119.387	199	1214
6	K-Means	20.140	41.620	489	3013
7	Random	36.127	88.802	291	1609
7	K-Means	18.706	38.790	501	3231
8	Random	35.284	82.967	343	1580
8	K-Means	17.592	36.676	664	3358

At $k = 4$, expected energy decreases from 162.505 mJ/round for random placement to 50.920 mJ/round for K-Means. The relative reduction is

$$\frac{162.505 - 50.920}{162.505} \times 100\% \approx 68.7\%. \tag{4.1}$$

The average sensor-to-gateway distance also decreases from 52.518 m to 23.560 m. The result is consistent with the distance-squared amplifier term in Equation 3.1: shorter links generally reduce transmission energy in this model. It should not be interpreted as a guarantee that K-Means will outperform every possible energy-aware or constrained placement algorithm.

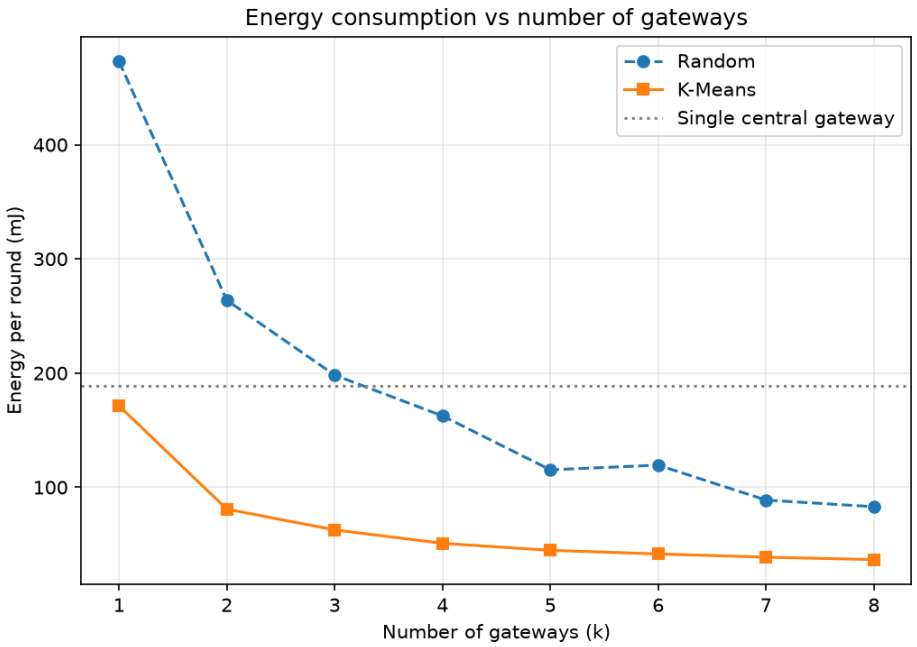


Figure 4.2: Expected sensor transmission energy per round versus gateway count.

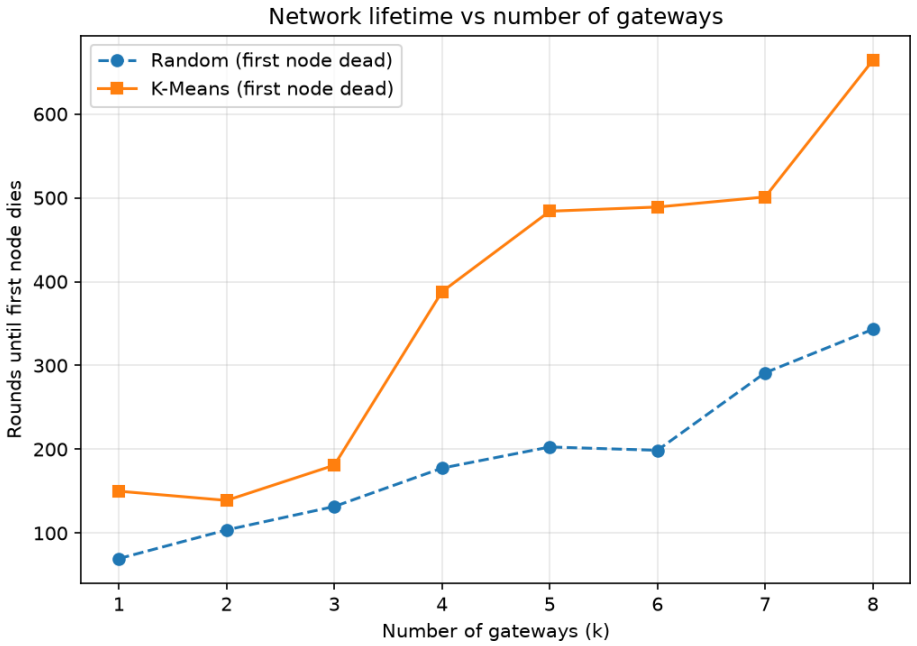


Figure 4.3: First-node-death round versus gateway count in the base experiment.

The K-Means lifetime values generally improve as gateway count increases in this run, because shorter sensor links reduce per-packet transmission energy. The relationship is not strictly monotonic for every random baseline metric because node placement and traffic are stochastic. The single central gateway baseline reported by the script has an average distance of 59.44 m and FND of 143 rounds; it is a reference point, not a range-constrained

coverage result.

4.2 Gateway-count selection

The extended script reports an elbow choice of $k = 3$, a maximum silhouette score at $k = 2$, and the minimum illustrative cost score at $k = 5$. Figure 4.4 shows the three criteria. These values differ because the criteria optimize different quantities. The cost minimum is conditional on the assumed gateway cost, battery cost, energy formula, and 5000-round horizon; it should not be called a universally optimal gateway count.

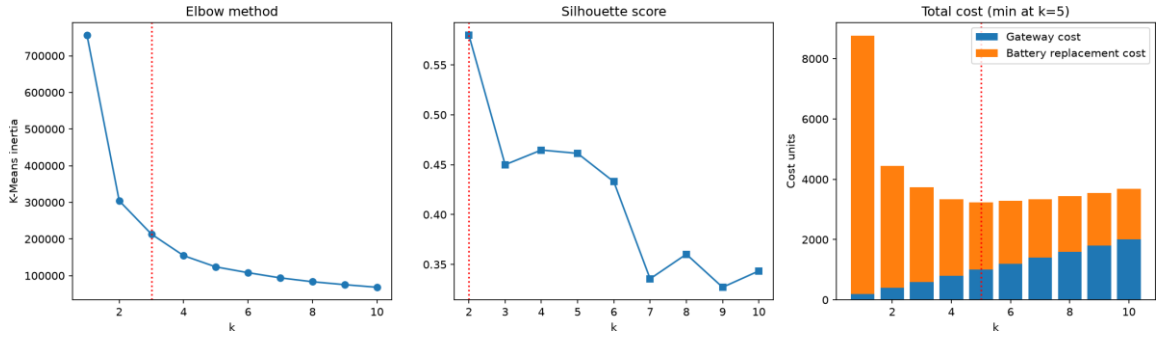


Figure 4.4: Gateway-count analysis using inertia, silhouette score, and the illustrative cost score.

4.3 Fault tolerance

At $k = 5$, the included output reports worst-case delivery after a single gateway failure of 68.0% for K-Means without backup, 74.0% for K-Means with backup selection, and 98.5% for K-Means with backup plus one grid-searched spare gateway. This indicates that the spare gateway improves in-range coverage under the model's failure rule. The result is not a measured packet delivery ratio and does not include real failure detection, interference, contention, or hardware failure modes.

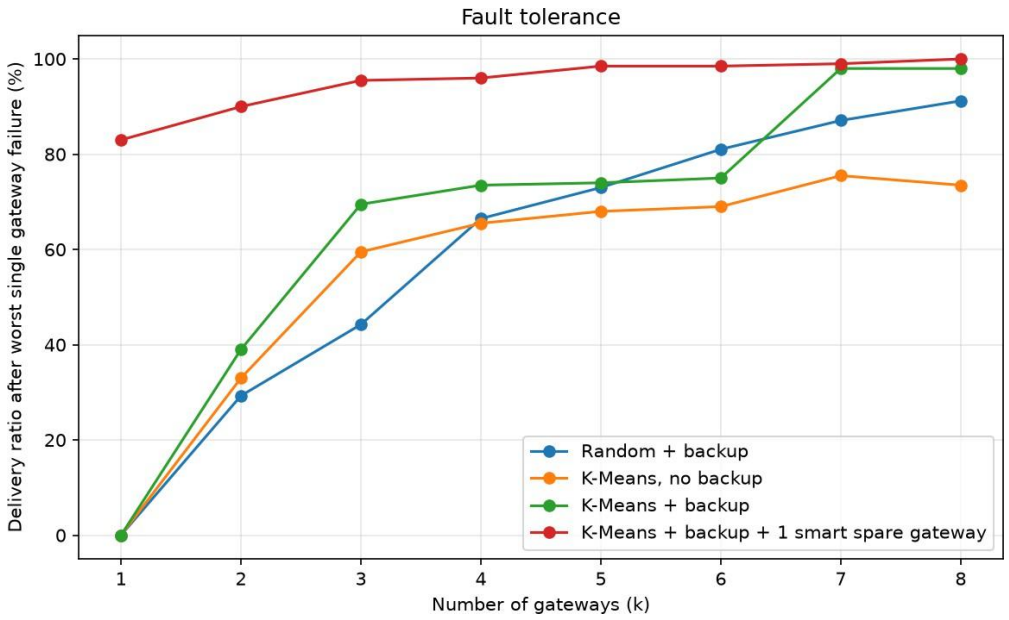


Figure 4.5: Worst-case delivery ratio proxy after a single gateway failure.

4.4 Multi-hop routing

The routing experiment uses $k = 2$ gateways and a direct range of 80 m with relay links limited to 40 m. Table 4.2 reproduces the principal values from the included extended-run log. The initial fraction of nodes able to reach a gateway is 95.5% for direct routing and 97.5% for both multi-hop modes. Residual-energy-aware multi-hop has the highest FND in this run, while its 50%-dead round remains lower than the direct mode. Therefore, no single lifetime metric should be used in isolation.

Table 4.2: Extended routing experiment ($k = 2$)

Routing mode	Initially reachable nodes (%)	FND (rounds)	50% dead (rounds)	Packets delivered
Direct (single hop)	95.5	370	2362	177282
Multi-hop (minimum energy)	97.5	279	1897	145470
Multi-hop (residual-energy aware)	97.5	785	1688	148546

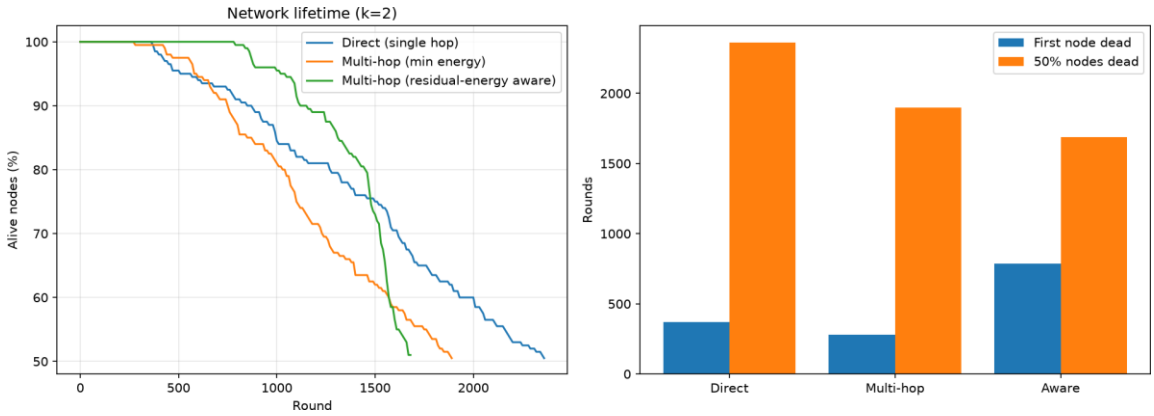


Figure 4.6: Alive-node history and lifetime metrics for the routing strategies.

The routing output illustrates a trade-off between delivery count and energy distribu-tion across relay nodes. Multi-hop routes can improve reachability where direct links are unavailable, but each relay consumes transmit and receive energy. The residual-energy weight changes the route cost and can delay the first depletion event in this specific run. Because the model uses one principal seed and simplified synchronous rounds, these observa-tions should be confirmed over multiple independent paired trials before drawing a general conclusion.

Chapter 5

Discussion and Conclusion

5.1 Discussion

The simulation demonstrates that gateway placement and gateway count significantly in-fluence sensor-to-gateway distance and energy consumption in Wireless Sensor Networks (WSNs). For the generated clustered deployment, K-Means placement achieves lower aver-age distance and expected transmission energy than random placement. At $k = 4$, it reduces expected transmission energy per round by approximately 68.7%. The gateway-count anal-ysis shows that the preferred value of k depends on the selection criterion, while backup and spare gateways improve in-range connectivity after a gateway failure.

The routing results indicate that direct and multi-hop strategies involve trade-offs. Residual-energy-aware routing increases the first-node-death (FND) round in the reported run, although packet delivery and overall lifetime may vary. These results depend on the topology, traffic seed, routing weights, and simulation assumptions.

5.2 Limitations and Future Work

The study uses a simplified radio-energy model and does not consider fading, interference, retransmissions, congestion, or complete gateway energy consumption. The base and ex-tended experiments use different communication-range assumptions. Furthermore, the re-ported results are simulation-based and have not been validated using physical hardware. The illustrative cost model does not represent actual market costs.

Future work should evaluate at least 20–30 independent deployment and traffic seeds, report statistical measures such as standard deviation and 95% confidence intervals, and apply consistent communication-range and energy-accounting assumptions. Comparisons with grid-based and energy-aware placement methods, ablation studies, and validation using

a network simulator or hardware testbed would strengthen the evaluation.

5.3 Conclusion

This project looked at a simple but important question in IoT design: where should gateways be placed so that sensor batteries last longer and the network keeps working when things go wrong? To answer it, a software-only simulation of 200 sensor nodes was built and tested in Python, along with a browser dashboard for live demonstration. In the base experiment, K-Means placement clearly beat random placement for every gateway count tested. With four gateways, it cut the estimated transmission energy per round by about 68.7% and pushed the first node death from round 178 to round 388. The cost analysis suggested that five gateways was the most economical choice for this layout, even though the elbow method and silhouette score preferred smaller values, because only the cost score takes the price of gateways and batteries into account.

The fault-tolerance results showed that a backup gateway alone helped only a little, raising worst-case delivery from 68.0% to 74.0%. Adding one carefully placed spare gateway lifted it to 98.5%, at the price of one more gateway. In the sparse two-gateway network, residual-energy-aware multi-hop routing delayed the first node death from round 351 to round 777, but it did not improve every measure. The 50%-node-death round and the number of delivered packets were lower than with direct routing, because relaying costs extra receive energy. These findings are limited to the models and assumptions used here, such as a single synthetic layout, no signal fading or packet loss, and cost values that are only illustrative. The framework is a reproducible starting point, but more independent experiments and stronger comparisons are needed before making general claims about performance or novelty.

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